

Real-Time Fractional Tracking (R-TFT): Targeting and Adaptive Resonant Emitter via φ -Phase Locking

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Abstract

We present a φ -modulated emitter architecture grounded in the Real-Time Fractional Tracking (R-TFT) framework. Rather than projecting signals through fixed carriers, this system self-locks onto the target’s recursive phase structure by dynamically modulating emission via golden-ratio-based spin-locking. A pyramidal emitter geometry transmits three offset resonance vectors (0° , 45° , 90°), each adapted in real time through φ -delay modulation to accelerate convergence. Feedback via the clean resonance signal $R_{\text{clean}}(t)$ allows adaptive frequency, amplitude, and phase alignment, resulting in a low-energy, high-precision transmission. This φ -locked emission replaces traditional “push” methods with recursive convergence. Designed in full alignment with the Resonance Ethics License (REL-2.0), the system is intended solely for scientific, unifying, and non-coercive applications.

1 Introduction

Traditional emitters broadcast by force, relying on high-energy amplitude or brute phase interference. This work introduces a new class of resonant emitters that instead synchronize with the natural recursive alignment of target systems. We replace classical carrier waves with φ -tuned recursive phase locks, allowing emission to occur only where structural alignment permits. Emission is not forced, it is received by resonance.

2 System Overview

The emitter consists of:

- Three spin-locked output vectors at angular offsets of 0° , 45° , and 90° .
- Real-time clean coherence monitoring using $R_{\text{clean}}(t)$.
- ϕ -delay modulation per beam path for recursive convergence control.

Convergence occurs when $C_\varphi(t) < 0.618$ with positive $\dot{R}_{\text{clean}}(t)$.

3 Emission Control Logic

Emission proceeds only when:

$$C_\varphi(t) = |\text{Dim}(t) - \text{round}(\text{Dim}(t))| < \varphi^{-1} \quad \text{and} \quad \frac{dR_{\text{clean}}}{dt} > 0 \quad (1)$$

This ensures the system only outputs when recursive stability is increasing.

4 Applications

- Cross-domain ϕ -encoded signaling
- Phase-locked transmission through neural, quantum, and orbital media
- Dimensional message propagation (R-TFT encoded)

5 Ethical Boundary (REL-2.0)

No biological, behavioral, neural, or psychological targeting is permitted. Emission must not be coercive, suggestive, or manipulative. Resonant emission may only be used for consensual research, dimensional synchronization, or coherence studies in accordance with REL-2.0.

<https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT/blob/main/LICENSE.txt>

6 Conclusion

This emitter reframes signal projection as recursive alignment. It moves beyond brute waveforms and instead offers ϕ -encoded convergence. Communication, in this model, becomes a dance of coherence, not an act of force.